

Adora PINN — Step 4: Spectrum from the initial model

As-built notes · Ivan Milic · 2026-09-20 · step 4 of pinn/plan.md

Step 4 synthesizes the Fe I 6301/6302 Stokes spectra from the **pretrained field**, exercising the full forward model that the inversion will differentiate. Files [pinn/rt_forward.py](#), [pinn/step4_spectrum.ipynb](#).

Forward model. Evaluate $f_\theta(x, y, z)$ on the $128 \times 128 \times 56$ grid; convert its outputs to RT inputs — gas pressure $P \rightarrow \text{EOS} \rightarrow n_e, n_{\text{H,tot}}$; Cartesian $B \rightarrow (|B|, \gamma, \chi)$ for a vertical LOS (`b_to_rt_angles`); v_z, v_{turb} pass through — and run the piecewise-linear polarised RT via `synth_cube_map` (EOS/pg path). `rt_forward.synth_field` wraps this and is reused by steps 5–6.

GPU. New jax-free helper [pinn/gpu.py](#): `use_gpu(device=None)` picks the idle GPU (`nvidia-smi`), sets `CUDA_VISIBLE_DEVICES` and disables XLA pre-allocation (polite on the shared 4×H100 node); pass `device=3` to force one. Called first in the notebook, before any jax import.

Results. Full-map synth on one H100 in **4 s** (was 85 s on CPU), Stokes (128, 128, 4, 201). $\max |V|/I_c = 0$ (the model is non-magnetic, $B=0$); the velocity structure imprints a center-of-gravity Doppler pattern that recovers the input v_z ; and the field spectrum matches a direct synth of the mock atmosphere to $\max |\Delta(I/I_c)| = 8.3 \times 10^{-3}$ — i.e. the pretraining accuracy — confirming the field is a faithful warm start for the RT.

Next. Step 5: a mock inversion testset from a prepared synthetic cube. Files uncommitted.